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# Time & Work

Q: A can do a piece of work in 10 days & B can do the same piece of work in 15 days. find how many days it would take for work to finish if A & B work together?

Ans. A — 10 days  Piece of work = LCM(10, 15)
B — 15 days  ⇒ 30

A( 1 day → $\frac{1}{10}$ days.

B ( 1 days → $\frac{1}{15}$ days)

| | |
|---|---|
| 5 | 10, 15 |
| 2 | 5, 3 |
| 3 | 2, 3 |
| | 1, 1 |

$A + B = \frac{1}{10} + \frac{1}{15} \Rightarrow \frac{1}{6}$

1 day → $\frac{1}{6}$ glass

6 day → 1 glass

A → $\frac{30}{10}$ → 3 chair

or

30 chair → 10 days

(3 chair) → 1 day

30 chair → 15 day

$\frac{30}{15}$ = 2 chair → 1 day

Total chair → A + B

= 3 + 2

→ 5 chair

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Q: A & B together can do a piece of work in 15 days & B alone can do same work in 20 days, find no. of days that A will take to do same work alone?

A:

LCM ⇒ (A, B) ⇒ (15, 20) = 60

A+B → 15 day → 60
1 day → 4

| | |
|---|---|
| 5 | 15, 20 |
| 4 | 3, 4 |
| 3 | 1 |

B → 20 days → 60
1 day → 3

So, A is doing 1 day → 1 work
so 60 day → 60 work
Ans

Q: A alone can do half work in 5 days. B alone can do 3/5 of same work in 9 days. C alone can do 2/3 of same work in 8 days. find no. of days it will take for work to be completed if all three work together?

A: A → 10 days
B → $\frac{3}{5}$ → 9 ⇒ 1 → 15 days
C → $\frac{2}{3}$ → 8 → 12 days

LCM ⇒ (10, 15, 12) = 60

| | |
|---|---|
| 5 | 10, 15, 12 |
| 2 | 1, 3, 12 |
| 3 | 1, 3, 6 |
| 4 | 1, 1, 2 |
| | 1, 1, 1 |

A → 1 day → 6
B → 1 day → 4
C → 1 day → 5
→ 15 → it will take 4

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Q: A & B can do a piece of work = 12 days. B & C can do it in 15 day, C & A in 20 days. A can do work alone in?

Ans

A+B → 12 → 5
B+C → 15 → 60 → 4
A+C → 20 → 3

LCM → 12, 15, 20 → 60

2(A+B+C) → 12

$\underbrace{A+B}_{}+\underbrace{C}_{15}$ → 12/2 = 6

A = 6 − 4

A → 2

If we have to 60 then $\frac{60}{2}$ = 30 day A need.

Q: A takes 4 days to complete 1/3 of job. B takes 3 days to complete 1/6 th of same work & C takes 5 days to complete 1/2 the job. If all of them work together for 3 days, A & C quit. How long it take for B to complete remaining work alone?

Ans:

A → 4 day
B → 18 day
C → 10 day

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$$\frac{A \text{ actual work}}{A \text{ alone work}} + \frac{B''}{B''} + \frac{C''}{C''} = 1$$

$$\frac{3}{12} + \frac{x}{18} + \frac{3}{10} = 1$$

$x = 8.1$ days

To complete remain work alone By b is?

8.1 day − 3 ⇒ 5.1 day

3 दिन काम किया